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AI3403: Multi-Agent Systems (MAS) - Assignment #4

Rules for Submitting Assignment

- The assignment is worth 100 points. You must present the entire work leading to their solution and also attach the associated code.
- The assignment will be due by **23:59 on Friday, October 09, 2026** in the Google classroom. The deadline is strict and no extensions shall be awarded unless otherwise for unavoidable circumstances.
- Usage of AI tools like ChatGPT/Claude/Gemini is permitted and must be limited only to understand the concepts taught in the lecture. If you use them in the assignment, you should explicitly declare where and to what extent you have used it. Educate yourself enough to know and spot when the AI tool that you are using is giving you some nonsense!
- You should submit your assignment as **one** pdf document and the file should be named using roll number. For instance, if your roll number is **ai25btech11111**, then the pdf should be named as **ai25btech11111Assignment4.pdf**. Please also write your name in the first page of your assignment.
- Students are strongly encouraged to submit their work in a neatly typed LaTeX document with all supporting code attached to the same document. An example overleaf latex template can be found in this link: `https://www.overleaf.com/latex/templates/assignment-homework-solutions-template/nfjybpqbdrrn`. If typing is not possible, hand-written submissions will be considered as long as they are legibly written and scanned properly with all the codes attached into one pdf.
- Students can use any coding language of their preference such as Matlab, Python, Julia, C++, etc. All the associated codes for the assignment problems should be attached at the end of each respective problem in your assignment submission. You can also use Jupyter notebook, Google colab, etc to do your assignment and submit it as one pdf.
- You can use internet, any books, lecture notes, etc. Discussing with other students is perfectly fine and in fact is strongly encouraged. But plagiarism (copying solutions from others without putting your hard work) will never be tolerated.

Problem 1: Data-Driven Epidemic Modelling and Vaccination on Networks
Consider a population of N individuals connected through an undirected contact network $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with adjacency matrix $A \in \mathbb{R}^{N \times N}$. Each individual can be in one of the following three states: Susceptible, Infected, and Recovered. The disease evolution is assumed to approximately follow an SIR model $S \longrightarrow I \longrightarrow R$. You are provided with two synthetic datasets:

- `network.csv`: the adjacency matrix, or equivalently the edge list, of the contact network;
- `epidemic_data.csv`: the observed infection state of every individual over a sequence of days.

The true transmission rate β and recovery rate γ used to generate the data are **not provided**. Your objective is to reconstruct the epidemic from the data, estimate the unknown SIR parameters, simulate the epidemic, determine whether the observed spreading behaviour corresponds to an epidemic, and design a vaccination strategy based on the topology of the contact network.

1. From the individual-level data, compute at every day k, the quantities $S[k], I[k]$, and $R[k]$. Plot them on the same figure. From the data, report the following:
   (a) the maximum number of simultaneously infected individuals,
   (b) the day on which the infection reaches its maximum,
   (c) the final number of individuals who experienced infection,
   (d) the fraction of the population that experienced infection.
2. Consider the continuous-time SIR model. Assume that the sampling interval is $\Delta t = 1$ day. Using a forward-Euler approximation, we can discretize the recovery dynamics as

$$R[k+1] - R[k] \approx \gamma I[k] \Delta t.$$

   When $I[k] \neq 0$, an estimate of the recovery rate can be obtained as

$$\gamma_k = \frac{R[k+1] - R[k]}{I[k]\Delta t}.$$

   Use the above idea and the given data set to estimate γ. You may, compute the estimate as

$$\hat{\gamma} = \frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \frac{R[k+1] - R[k]}{I[k]\Delta t},$$

   where $\mathcal{K} = \{k : I[k] > 0\}$ denotes the days when the number of infected people was strictly positive. Using estimated value $\hat{\gamma}$, report the approximate average infectious duration.
3. Use a similar approach to discretize the equation for susceptible population and estimate the transmission rate β whenever $S[k]I[k] \neq 0$ and inferring the $\mathcal{K}_\beta = \{k : S[k]I[k] > 0\}$ (from the given data set) which denotes the days when $S[k]I[k] > 0$. Report the estimated

transmission rate $\hat{\beta}$. Using the estimates $\hat{\beta}$ and $\hat{\gamma}$, compute the reproduction number and interpret the obtained value as to whether the infection can initially grows or tends to die out. Also comment whether the observed disease has the potential to generate an epidemic.

4. Using parameters $\hat{\beta}$ and $\hat{\gamma}$, form the continuous time SIR model. Infer the initial values from the given data as $S(0) = S[0], I(0) = I[0], R(0) = R[0]$. Simulate the SIR model and plot the simulated trajectories of $S(t), I(t)$, and $R(t)$. Comment on how well the fitted SIR model explains the synthetic epidemic data.

5. You should now use both the data and the mathematical model to determine whether the observed infection constituted an epidemic. Your answer must be based on evidences like the observed infection trajectory $I[k]$, the estimated reproduction number $\hat{R}_0$, the magnitude of the infection peak, and the final fraction of the population infected.

6. Suppose that a vaccine is available before the epidemic begins, but only a limited number of individuals can be vaccinated. Assume that you are permitted to vaccinate at most $B = 15$ individuals. A vaccinated individual is assumed to be completely immune and does not participate in disease transmission. Your task is to determine which nodes in the network should be vaccinated in order to reduce the spread of infection. Compare the following vaccination strategies:

   (a) **Random vaccination:** Select B nodes uniformly at random.

   (b) **Degree-based vaccination:** Compute the degree $d_i = \sum_{j=1}^{N} a_{ij}$ of every node and vaccinate the B nodes having the largest degree.

   (c) **Eigenvector-centrality-based vaccination:** Compute the dominant eigenvector v satisfying $Av = \lambda_{\max}(A)v$. Vaccinate the B nodes corresponding to the largest entries of v.

   For every vaccination strategy, remove the vaccinated individuals from the susceptible population before the epidemic starts (that is at time $t = 0$) and simulate the epidemic again. Compare the following quantities:

$$I_{\max} = \max_k I[k], \quad T_{\text{peak}} = \arg\max_k I[k], \quad N_{\text{infected}} = N - S[\infty].$$

   Plot the infection trajectories for all vaccination strategies. Answer the following questions.

   (a) Which vaccination strategy performed best?

   (b) Why did vaccinating certain nodes have a larger effect than vaccinating others?

   (c) Is vaccinating the highest-degree nodes always the same as vaccinating the nodes with highest eigenvector centrality?

Problem 2: Temporal Logic Specification for Cyber-security System
Consider an online authentication system for a secure computing service. Each user account can experience login attempts, failed authentications, successful authentications, account locking, administrator verification, password-reset requests, user notifications, and account unlocking. The security system also continuously monitors quantities such as the user's login anomaly score, the rate of failed login attempts, and the confidence that the current user is legitimate. Your task is to express the security requirements given below using LTL, MTL, and STL. Simply writing a temporal-logic formula without explaining its meaning will not receive full credit.

Consider using the following Boolean atomic propositions:

`LoginFailure` :	the current login attempt has failed,
`LoginSuccess` :	the current login attempt is successful,
`AccountLocked` :	the account is locked,
`AccountUnlocked` :	the account is unlocked,
`AdminVerified` :	administrator verification has been completed,
`PasswordResetRequested` :	a password-reset request has been made,
`UserNotified` :	the user has been notified of a security event,
`SuspiciousActivityDetected` :	suspicious login behaviour has been detected.

Assume that the authentication system continuously monitors the following real-valued signals:

$\texttt{AnomalyScore}(t)$:	login anomaly score at time t,
$\texttt{FailedAttemptRate}(t)$:	number/rate of failed login attempts over a recent time window,
$\texttt{LegitimateUserConfidence}(t)$:	confidence that the current user is legitimate,
$\texttt{AccountLockStatus}(t)$:	account-lock status at time t.

Assume that

$$\texttt{AccountLockStatus}(t) = \begin{cases} 1, & \text{the account is locked,} \\ 0, & \text{the account is unlocked} \end{cases}.$$

Larger values of $\texttt{AnomalyScore}(t)$ and $\texttt{FailedAttemptRate}(t)$ indicate more suspicious behaviour, whereas larger values of $\texttt{LegitimateUserConfidence}(t)$ indicate greater confidence that the user is legitimate. Using LTL, encode the following security requirements. For each specification, write the corresponding LTL formula and briefly explain why your formula represents the given requirement.

A1. Whenever an account is locked, a successful login must not occur.

A2. Once the account becomes locked, it should remain locked until administrator verification has been completed.

A3. Whenever administrator verification is completed for a locked account, the account must eventually become unlocked.

A4. Whenever a password-reset request is received, the account must eventually either become unlocked or administrator verification must be completed.

A5. A successful login and a failed login attempt must never occur simultaneously.

A6. Whenever an account becomes locked, the corresponding user must eventually be notified of the security event.

Encode each of the following requirements using MTL. For each requirement, write the corresponding MTL formula; and explain briefly how the time interval appearing in the formula represents the stated timing requirement.

B1. Whenever `SuspiciousActivityDetected` becomes true, the account must become locked within 10 seconds.

B2. Whenever the account becomes locked, the user must be notified within 30 seconds.

B3. Whenever a password-reset request is made, administrator verification must be completed within 10 minutes.

B4. Once an account becomes locked, it must remain locked for at least 60 seconds unless administrator verification has already been completed.

B5. Once administrator verification has been successfully completed, the account must become unlocked within 2 minutes.

B6. Whenever a failed-login event occurs, then within the following 20 seconds at least one of the following must happen: 1) a legitimate login succeeds, or 2) the account becomes locked.

Suppose that the security system now uses continuous-valued monitoring signals. Assume the following security thresholds $A_{\text{crit}} = 0.8$, $F_{\text{crit}} = 5$, $C_{\text{safe}} = 0.9$. Encode the following requirements using STL. For each STL requirement, identify the atomic signal predicates, write the corresponding STL formula, and explain the physical meaning of the formula.

C1. During normal operation, the login anomaly score should always remain below the critical value meaning that $\texttt{AnomalyScore}(t) < A_{\text{crit}}$. Express this requirement using STL.

C2. Whenever $\texttt{AnomalyScore}(t) > A_{\text{crit}}$, the account must become locked within 10 seconds.

C3. Whenever $\texttt{FailedAttemptRate}(t) > F_{\text{crit}}$, the account must become locked within 5 seconds.

C4. Suppose an account is currently locked. Whenever both $\texttt{LegitimateUserConfidence}(t) > C_{\text{safe}}$ and $\texttt{AnomalyScore}(t) < 0.3$ hold, the account should become unlocked within 2 minutes.

C5. Whenever $\texttt{AnomalyScore}(t) > 0.8$ and $\texttt{LegitimateUserConfidence}(t) < 0.5$, the account

must remain locked.

C6. Suppose that at some instant, $\texttt{AnomalyScore}(t) > 0.8$ and continues to remain above 0.8 for at least 5 seconds. Express the requirement that such persistent suspicious behaviour must cause the account to become locked within the following 5 seconds.

Answer the following conceptual questions.

D1. Why is MTL more appropriate than LTL for a requirement such as

"The account must be locked within 10 seconds after suspicious activity is detected."

D2. Why is STL more appropriate when security requirements involve real-valued signals such as

$$\texttt{AnomalyScore}(t), \qquad \texttt{FailedAttemptRate}(t), \qquad \texttt{LegitimateUserConfidence}(t)?$$

D3. Explain the conceptual difference between the following three specifications:

$$\square\left(\texttt{SuspiciousActivityDetected} \rightarrow \lozenge \texttt{AccountLocked}\right),$$

$$\square\Big(\texttt{SuspiciousActivityDetected} \rightarrow \lozenge_{[0,10]} \texttt{AccountLocked}\Big),$$

and

$$\square\Big(\Big(\texttt{AnomalyScore}(t) > 0.8\Big) \rightarrow \lozenge_{[0,10]}\Big(\texttt{AccountLockStatus}(t) = 1\Big)\Big).$$

Clearly identify which of these formulas belongs to LTL, MTL, and STL, and explain what additional expressive capability is introduced when moving from one logic to the next.

Construct examples satisfying the following requirements.

E1. Construct one Boolean event trace that satisfies all the LTL specifications in Part A.

E2. Construct one Boolean event trace that violates at least two of the LTL specifications in Part A. For each violation, identify the specification and explain why the trace violates it.

E3. Construct one timed event trace that satisfies all the MTL specifications in Part B. The trace should explicitly indicate the times at which events occur.

E4. Construct one timed trace in which `SuspiciousActivityDetected` occurs, but the requirement "the account must become locked within 10 seconds" is violated. Clearly indicate:

- the time at which suspicious activity is detected;
- the time at which the account becomes locked, if it does;
- the MTL specification that is violated.